TORGIMA SYSTEM INTEGRITY AUDIT

Audit date: 2026-05-27

Scope: Full read of repo structure, engines, workers, routes, schema, and execution-loop tracing (static analysis + existing completeSystemDiagnostic.ts + targeted file reads).

Note: The audit template referenced Prisma/Next.js App Router paths. The actual stack is documented in docs/AUDIT_LOG.md and TORGIMA_COMPLETE_DIAGNOSTIC_REPORT.md.

Stack mapping (template vs actual)

```
Template expectation  |  Actual repository
Next.js 14 App Router + src/app/api/  |  Vite/React SPA (artifacts/torgima) + Express 5 API (artifacts/api-server/src/routes/)
Prisma + prisma/schema.prisma  |  Drizzle (lib/db/src/schema/*.ts)
src/lib/prisma.ts  |  @workspace/db pool + Drizzle (lib/db/src/index.ts)
src/bhudhi/*  |  artifacts/api-server/src/bhudhi/*
src/engines/execution-loop/*Engine.ts  |  engines/feedback, signal, condition, control, task-generation, task-guarantee
NextAuth.js v5  |  Custom magic-link (AuthService, sessions, cookie torgima_session)
42 tables in Prisma  |  348 pgTable definitions; Phase-1 core ~46 tables in completeSystemDiagnostic REQUIRED_TABLES
```

CRITICAL BREAKS (execution loop cannot function)

```
#  |  File  |  Exact issue  |  Law / impact
C1  |  artifacts/api-server/src/routes/feedback.ts:308-327  |  POST /api/feedback/task calls FeedbackService.submit only — no FEEDBACK_SUBMITTED / outbox event  |  Legacy clients never enter feedback!’signal!’condition!’enforcement!’task chain
C2  |  Production DB (operational)  |  bhudhi_logs had no CREATE TABLE in manual migrations until manual-20260601-bhudhi-log-repair.sql; if not applied, Bhudhi logging fails (code now fail-open in BhudhiCore.ts)  |  Usage caps & audit trail broken; AI may run without logs
C3  |  Production DB (operational)  |  bhudhi_signals ALTER in manual-20260528-structured-feedback.sql fails if table missing; repair in manual-20260601-missing-tables-repair.sql must be applied  |  Structured feedback / signal persistence warnings
C4  |  docs/AUDIT_LOG.md:64  |  scripts/apply-partial-indexes.sql exists but is manual — if not run on Supabase, is_current uniqueness is not enforced  |  Race / duplicate current rows possible under concurrency
```

Note: The primary production path POST /api/orders/:id/feedback !’ processFeedback() !’ outbox is CONNECTED. C1 is critical only for clients still on the legacy route.

BROKEN CHAINS (events not connected)

```
Handoff  |  Status  |  Evidence
POST /api/orders/:id/feedback !’ full loop  |  CONNECTED  |  orders.ts:346-364, FeedbackProcessingEngine.ts:305-326, eventContracts.ts, workers/start.ts
POST /api/feedback/submit (structured) !’ FeedbackWorker  |  PARTIAL  |  StructuredFeedbackService.ts:365-378; FeedbackWorker skips psychology/friction without feedback_id; SIGNAL_RECALCULATION still emitted
POST /api/feedback/task (legacy)  |  BROKEN  |  feedback.ts:308-327 — no event emission
feedback.ts weekly review / phase checkpoint  |  BROKEN (shortcut)  |  Emits task_generation_requested directly — bypasses signal!’condition!’control
```

```
EnforcementPackageGenerator in loop  |  N/A (stub)  |  Live path uses
ControlEngine.generateEnforcementPackage
```

ARCHITECTURAL VIOLATIONS

```
File / area  |  Violation  |  Per spec  |  Actual
Repo root  |  Stack mismatch  |  Prisma + Next App Router  |  Drizzle + Express + Vite
prisma/schema.prisma, src/app/api/**  |  Missing  |  Required  |  Does not exist
~40 route modules  |  Direct DB in routes  |  Services only  |  @workspace/db inline queries
feedback.ts, internal.ts  |  Bhudhi from routes  |  Engines only  |  routeBhudhiRequest from
TriggerRouter
TriggerRouter.ts  |  Anthropic SDK  |  Only BhudhiCore  |  By design (B1): sole
messages.create site
ErectorOrchestrationService.ts  |  console.log  |  logger  |  2× console.log in engines
boot.ts, queue.ts  |  console  |  logger  |  Startup / pg-boss paths
Auth  |  NextAuth  |  NextAuth v5  |  AuthService + sessions
MemoryManager.ts  |  Missing  |  Tier depth  |  BhudhiMemoryService, memoryPolicy,
ContextBuilder
bhudhi/prompts/  |  Missing dir  |  Prompt files  |  PromptRegistry, FallbackLibrary
MilestoneWorker, SystemHealthWorker  |  Missing names  |  Workers  |  RewardWorker,
HealthWorker
npm run lint  |  Missing script  |  Zero lint errors  |  Not in root package.json
```

MISSING FILES (template paths)

- prisma/schema.prisma !’ lib/db/src/schema/tables.ts (+ siblings)
- src/lib/prisma.ts !’ lib/db/src/index.ts
- src/lib/queue.ts !’ artifacts/api-server/src/lib/queue.ts
- src/lib/errors.ts !’ artifacts/api-server/src/lib/errors.ts
- src/lib/logger.ts !’ artifacts/api-server/src/lib/logger.ts
- src/lib/anthropic.ts !’ artifacts/api-server/src/lib/anthropic.ts
- src/types/api.ts !’ artifacts/api-server/src/types/api.ts
- src/app/api/** !’ artifacts/api-server/src/routes/**
- execution-loop/*Engine.ts paths !’ engines/feedback, signal, condition, control, task-generation, task-guarantee
- MemoryManager.ts !’ BhudhiMemoryService.ts
- prisma/seed.ts !’ lib/db/src/seed.ts
- MilestoneWorker.ts !’ RewardWorker.ts
- SystemHealthWorker.ts !’ HealthWorker.ts

SCHEMA ISSUES

- 348 Drizzle tables total; Phase-1 core ~46 in REQUIRED_TABLES (0 missing in schema per static audit).
- bhudhi_logs: repair in manual-20260601-bhudhi-log-repair.sql — must be applied on production.
- bhudhi_signals: repair in manual-20260601-missing-tables-repair.sql.
- order_status.queued: repair in manual-20260601-missing-tables-repair.sql.
- scripts/apply-partial-indexes.sql: all 6 indexes present in repo; DB application not verified in static audit.

TYPE ERRORS

- pnpm typecheck passes on current main.
- No project-wide npm run lint — criterion not measurable without adding ESLint.

WARNINGS (non-critical)

1. EnforcementPackageGenerator.ts — stub/TODO
2. Structured feedback: FeedbackWorker skips branches without feedback_id
3. feedback.ts weekly/phase paths skip full loop
4. state_version bumps in workers, not all engines directly
5. Mix of success()/failure() vs raw res.json in some routes
6. .env.example gaps (NEXTAUTH vs SESSION_SECRET)
7. Bhudhi memory tier labels not explicit session/7d/30d
8. console.log in Erector / seed scripts
9. Production env keys may be unset locally
10. db:apply-manual + apply-partial-indexes.sql required per environment

---

VERIFIED WORKING

- TorgimaError, ApiResponse<T>, pino logger, pg-boss singleton
- WorkerRegistry + 31 workers via workers/start.ts
- BaseWorkerHandler: idempotency, event_logs, dead_letter_events
- Primary execution loop event map (eventContracts.ts)
- is_current transactional flips in Signal, Condition, Control engines
- TaskGenerationEngine reads enforcement_packages (throws if missing)
- TriggerRouter: checkTriggerAllowed, checkUsageAllowed
- OutputValidator in BhudhiCore; fallbacks via FallbackLibrary
- Anthropic SDK only in TriggerRouter.ts
- Partial indexes script complete (6 indexes)
- Feature flags seed in lib/db/src/seed.ts
- POST /api/v1/onboarding/select-blueprint launch route

COMPLETION CRITERIA SCORECARD

```
#  |  Criterion  |  Result
1  |  tsc --noEmit !’ 0 errors  |  PASS (pnpm typecheck)
2  |  npm run lint !’ 0 errors  |  FAIL — no lint script
3  |  Execution loop traceable  |  PASS primary; FAIL legacy/shortcuts
4  |  42 tables in prisma/schema  |  N/A — Drizzle, ~46 core OK
5  |  apply-partial-indexes.sql (6)  |  PASS file; UNKNOWN DB
6  |  No route direct DB  |  FAIL — widespread
7  |  No route direct Anthropic  |  PASS
8  |  Enforcement null-check  |  PASS TaskGenerationEngine
```

```
9  |  Bhudhi fallbacks  |  PASS
10  |  WorkerRegistry complete  |  PASS
```

TOTALS

```
Category  |  Count
Critical breaks  |  4
Broken / partial chains  |  3
Architectural violations  |  15+
Missing files (template paths)  |  18 path groups
Schema issues  |  5
Type errors (compile)  |  0
Warnings  |  10
```

Recommended Phase 3 fix order

1. P1: Wire or remove POST /api/feedback/task; fix weekly/phase shortcuts.
2. P2: Run pnpm db:apply-manual + apply-partial-indexes.sql on production.
3. P3: Move route-level Drizzle to services.
4. P4: Bhudhi log monitoring; structured feedback payload alignment.
5. P5: Worker audit (already strong); add lint CI.
6. P6: Add pnpm lint and enforce zero errors.

Generated from Torgima Execution OS codebase audit — Phase 2 report only; repairs not included.